Artur Iasenovets, M.S. candidate

Present Position	**Graduate Student Researcher** Chongqing University of Posts and Telecommunications	*Email:* L202420002@stu.cqupt.edu.cn *Publication list:* ORCID

Research Interests

My interests lie in **applied cryptography** and **privacy-enhancing technologies**, with a focus on **homomorphic encryption**, **private information retrieval**, **searchable encryption**, and privacy-preserving systems. I am particularly interested in practical cryptographic protocols for private data search and record retrieval.

Additional Information

Research profiles: Portfolio • LinkedIn
References: Available upon request.

Education

Chongqing University of Posts and Telecommunications, Chongqing, China
School of Cyberspace Security and Information Law
- **M.S.** Network & Information Security — Sep. 2024–Jul. 2027 *(expected)*
  - Supervisor: Prof. Tang Fei
  - GPA: 3.98/4.00

Ukhta State Technical University, Ukhta, Russia
- **B.S.** Information Systems and Technologies — Mar. 2022–Jun. 2024
  - GPA: 4.46/5.00

St. Petersburg State University of Telecommunications, St. Petersburg, Russia
- **B.S. studies** in Information Systems and Technologies — Sep. 2020–Mar. 2022

Publications

Publications and Preprints

Artur Iasenovets, Fei Tang, H. Zhu, P. Wang, and L. Liu. "Bringing Private Reads to Hyperledger Fabric via Private Information Retrieval," arXiv:2511.02656, 2025. Manuscript under review.

H. Zhu, F. Tang, W. Qin, J. Shan, P. Wang, and Artur Iasenovets. "EVMK-SSE: Efficient and Verifiable Multi-Keyword Symmetric Searchable Encryption," *Journal of King Saud University – Computer and Information Sciences*, 2025.

Artur Iasenovets and Fei Tang. "MFCTIIF: Multi-Feed Cyber Threat Intelligence Integration Framework," *International Research Journal*, no. 1 (163), 2026.

Manuscript in Preparation

Piyumi M. S. Rajapaksha*, Artur Iasenovets*, Yinxue Yi, and Fei Tang. "TWAPPA-FRL: Trust-Weighted Privacy-Preserving Aggregation for Federated Reinforcement Learning in Collaborative Intrusion Detection," manuscript in preparation, 2026. *Equal contribution.

Awards & Achievements

Chinese Government Scholarship for Graduate Studies — 2024–2027
Full scholarship for the M.S. programme at CQUPT.

Winner, YouKnow EdTech Project Competition — 2023
Won the EdTech project competition organized by the MIPT Phystech School, the Fund for Development of Phystech Schools, and Startech.Base.

First Degree Diploma, Severgeoecotech — 2022
Awarded first place in the Modern Information Technologies section of the 23rd International Youth Scientific Conference.

Sber Student Accelerator Regional Demo Day — 2023
Completed the Sber Student Accelerator with the AcademAI team and participated in the North-Western regional demo day.

Supporting certificates

Research Experience

Graduate Student Researcher @ Cryptography and Applications Research Group, CQUPT — Sep. 2024–Present

Private Information Retrieval for Hyperledger Fabric

- Designed and implemented a BGV-based computational PIR protocol in Go using Lattigo, integrating private world-state reads directly into Hyperledger Fabric chaincode.
- Defined the honest-but-curious (HBC) threat model and analysed query privacy under the IND-CPA security of BGV, including repeated-query security and observable leakage.
- Built the experimental workflow and evaluated correctness, latency, communication, storage, and deployment overhead across several cryptographic parameter sets.

Privacy-Preserving Aggregation for Federated Reinforcement Learning

- Co-developed TWAPPA-FRL, combining trust-weighted aggregation (TWA) with CKKS-based privacy-preserving aggregation (PPA) for collaborative intrusion detection, as an equal-contribution author.
- Combined trust-aware client weighting with CKKS-based encrypted aggregation of federated model updates and aggregate-only decryption by a separate non-training key authority.
- Developed the threat model and security analysis, including individual-update confidentiality, explicit leakage, and the assumptions and limitations of the PPA-enabled paths.

Verifiable Multi-Keyword Searchable Encryption

- Contributed to EVMK-SSE, an efficient multi-keyword searchable encryption scheme based on an OT-based client-independent relaxed OPRF.
- Reviewed the security analysis covering query privacy, explicit search and access-pattern leakage, and result verifiability against a malicious cloud server.

Industry Experience

Security Intern @ Positive Technologies — Aug. 2024–Feb. 2025

- Developed tested Python security tooling for the MaxPatrol EDR team to validate Windows file paths, network configurations, and endpoint-control policies.
- Built an Airflow-based malware-processing pipeline that collected samples from VX-Underground, stored artifacts in S3/MinIO, and performed YARA-based scanning.
- Investigated exploitation and post-exploitation activity involving CVE-2020-5902 and CVE-2023-38831 through EVTX analysis, PowerShell deobfuscation, IoC extraction, and MITRE ATT&CK mapping.
- Extended this threat-intelligence experience into the first-author MFCTIIF research project on multi-feed CTI integration.

Software Engineer @ AcademAI — Aug. 2023–Aug. 2024

- Co-developed an end-to-end AI platform for generating structured and personalized educational courses from user-defined topics and learning objectives.
- Implemented a Python REST API and LangChain-based generation pipeline using structured prompting and asynchronous LLM orchestration.
- Built the frontend and application layer using Next.js, TypeScript, Prisma, and Auth.js, including authentication and persistent course data.
- Containerized and deployed the platform using Docker and GitHub Actions.

Skills

- ◇ **Go, Python, TypeScript:** proficient
- ◇ **C++, Rust:** familiar
- ◇ **Homomorphic encryption:** BGV, BFV, CKKS
- ◇ **Private search:** PIR/CPIR, searchable encryption
- ◇ **HE libraries:** Lattigo, Microsoft SEAL, OpenFHE, TenSEAL
- ◇ **Systems:** Hyperledger Fabric, Linux, Docker, Git
- ◇ **Security engineering:** YARA, Chainsaw, malware and log analysis
- ◇ **Languages:** Russian (native), English (C1), Chinese (HSK 3)